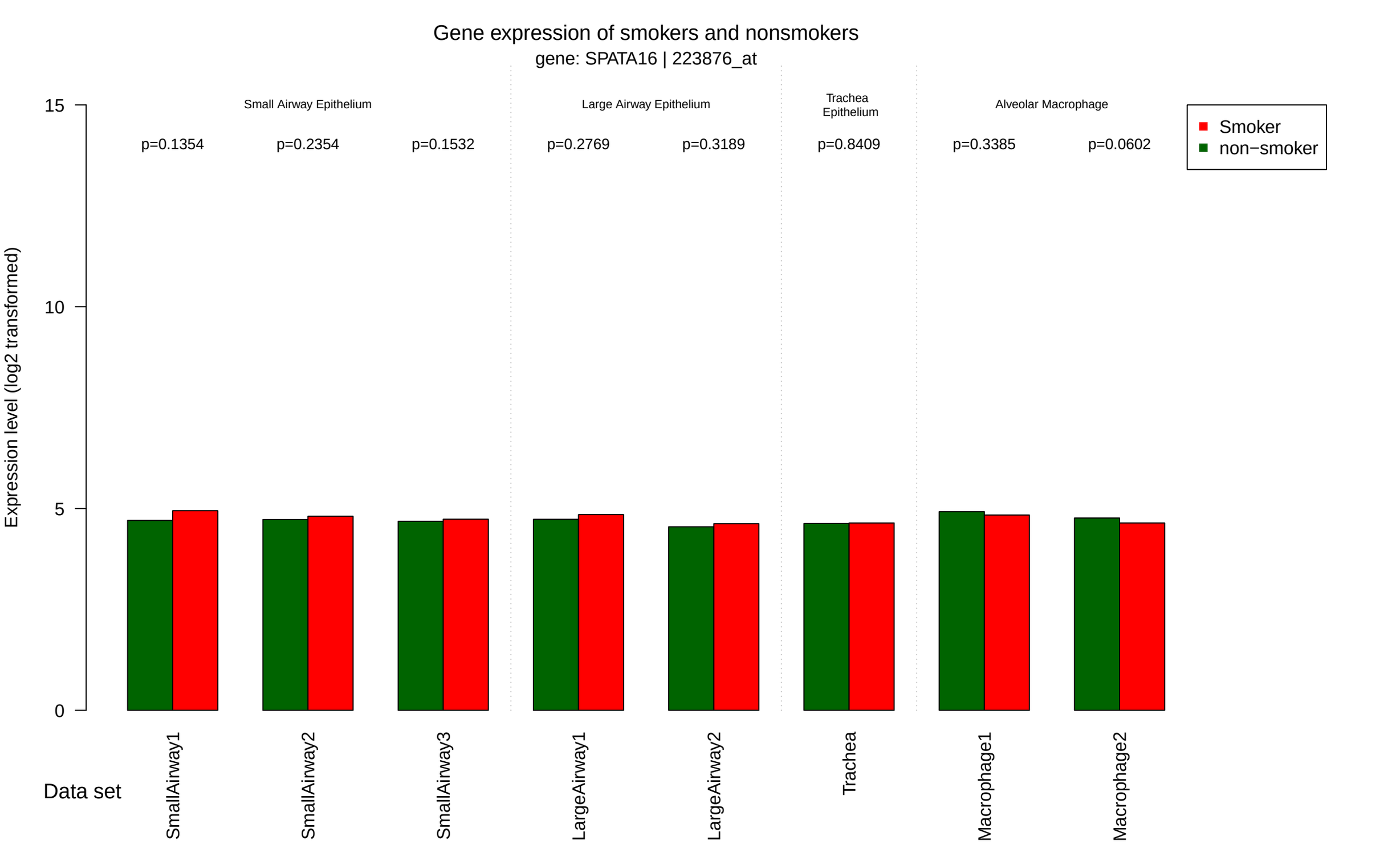
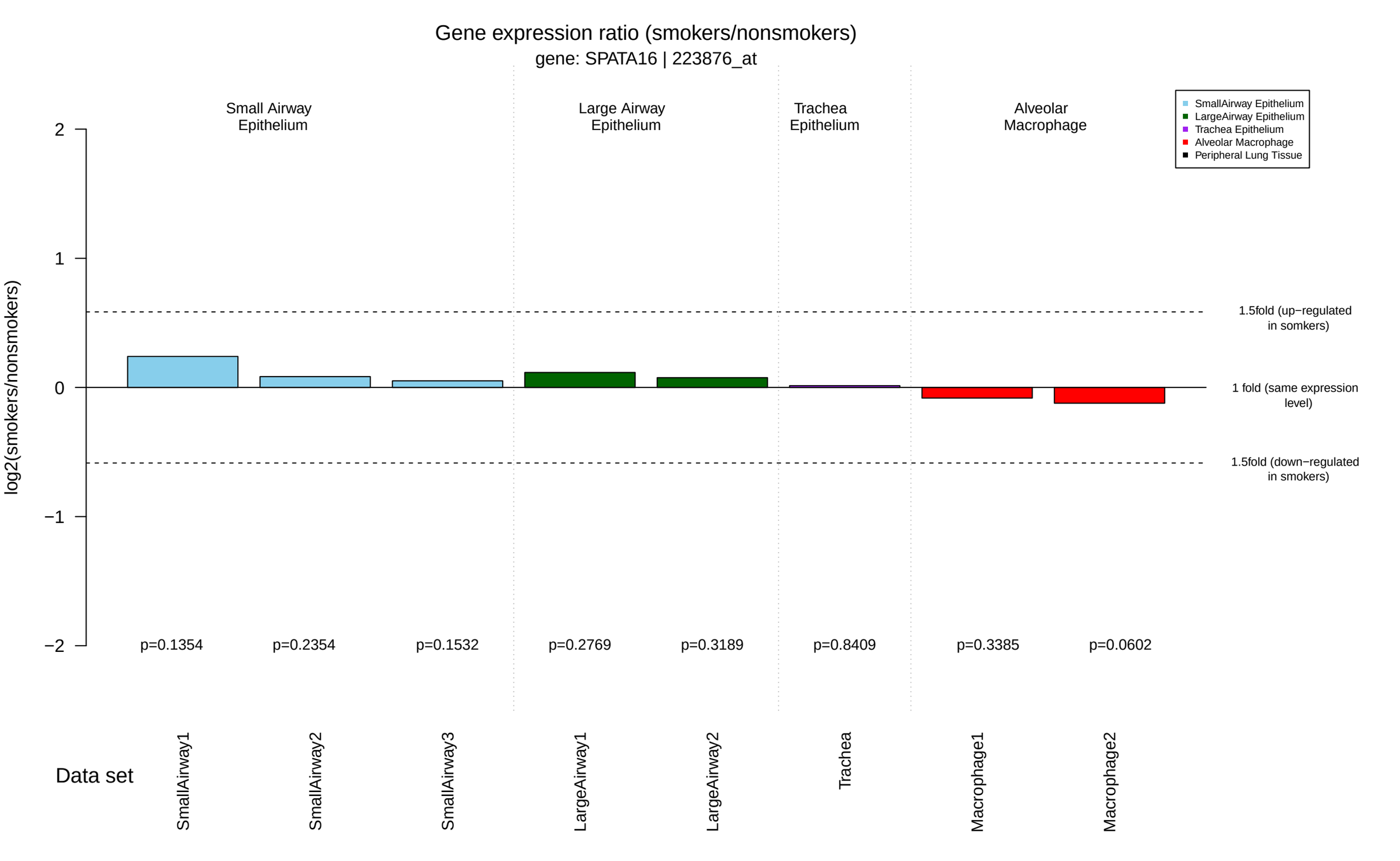
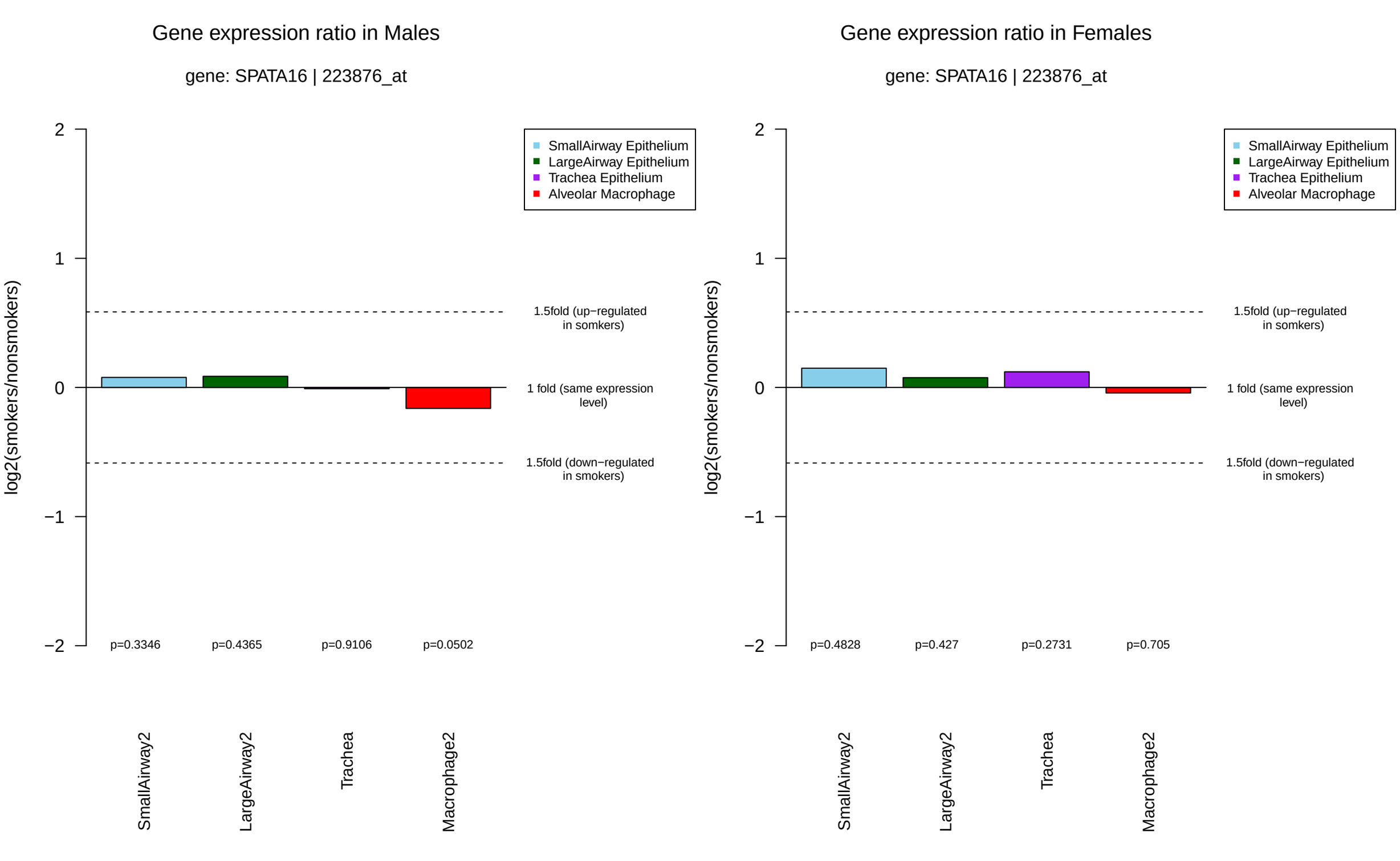
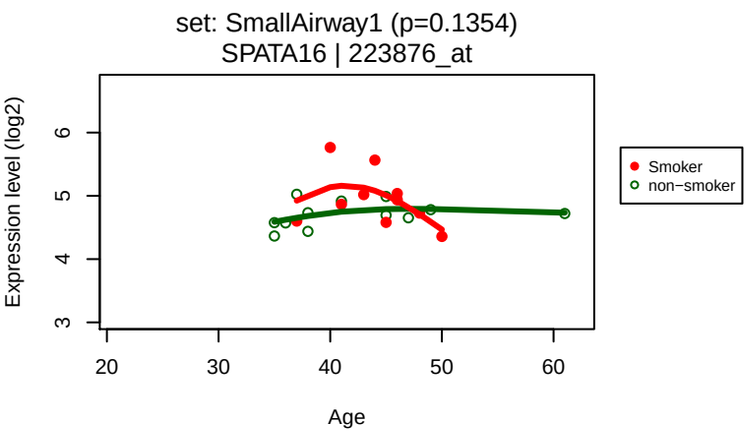
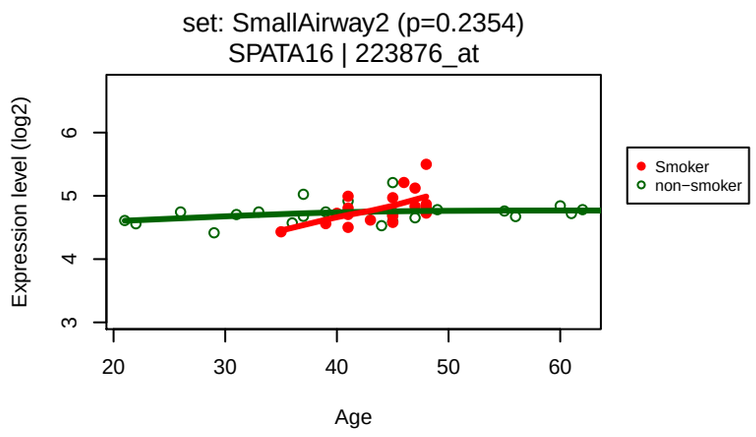
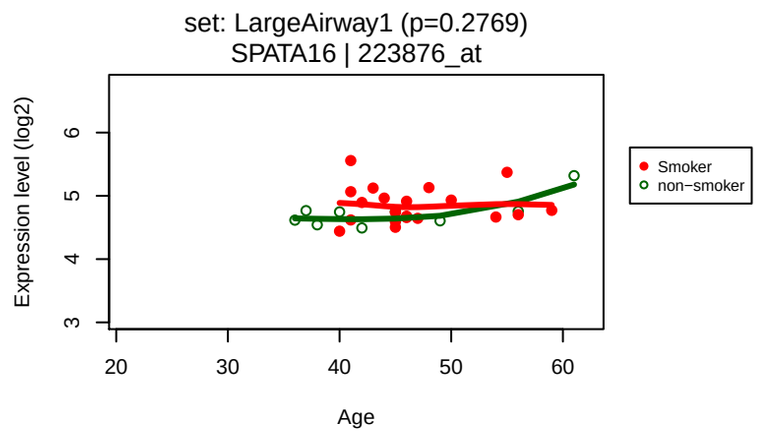

Gene expression of smokers and nonsmokers
gene: SPATA16 | 223876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1354
p=0.2354
p=0.1532
p=0.2769
p=0.3189
p=0.8409
p=0.3385
p=0.0602
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

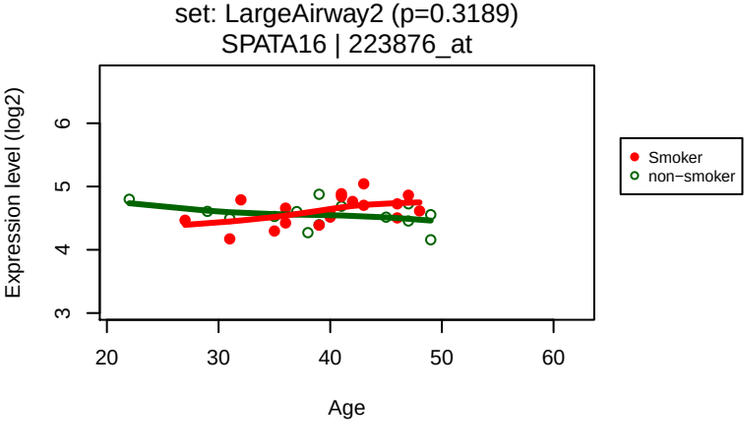

Gene expression ratio (smokers/nonsmokers)
gene: SPATA16 | 223876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1354
p=0.2354
p=0.1532
p=0.2769
p=0.3189
p=0.8409
p=0.3385
p=0.0602
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

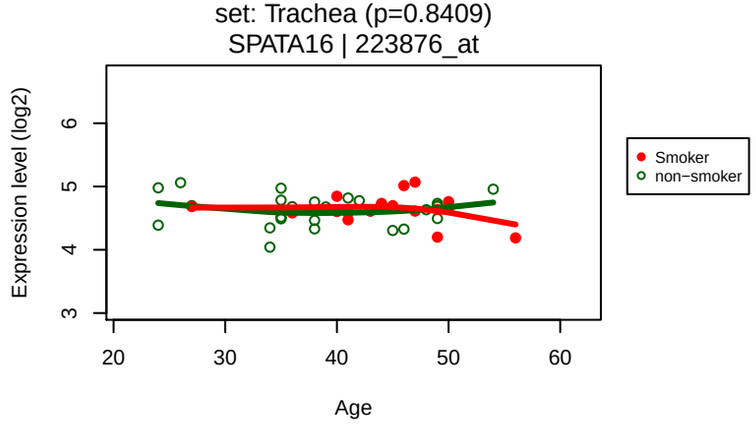

Gene expression ratio in Males
gene: SPATA16 | 223876_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3346
p=0.4365
p=0.9106
p=0.0502
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA16 | 223876_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4828
p=0.427
p=0.2731
p=0.705
SmallAirway2
LargeAirway2
Trachea
Macrophage2

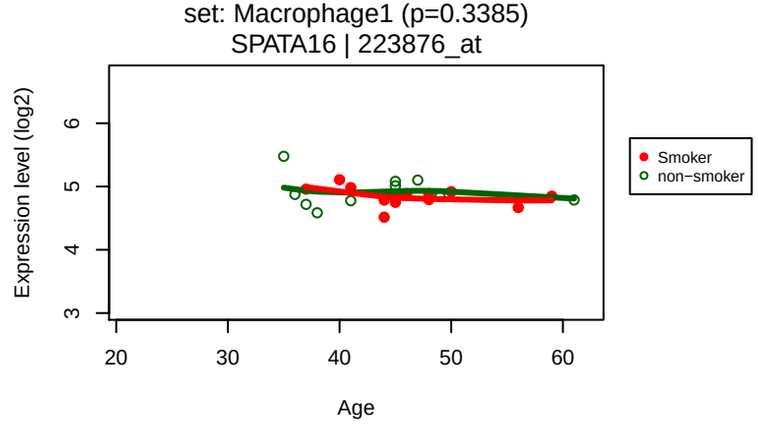

set: SmallAirway1 (p=0.1354)
SPATA16 | 223876_at
Expression level (log2)
Age
Smoker
non−smoker

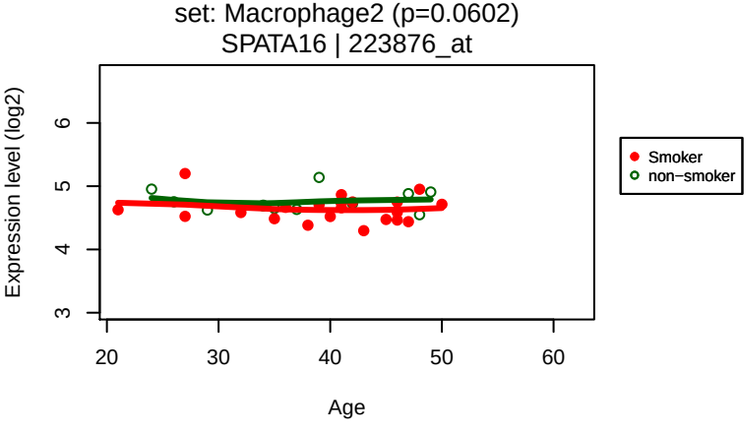

set: SmallAirway2 (p=0.2354)
SPATA16 | 223876_at
Expression level (log2)
Age
Smoker
non−smoker

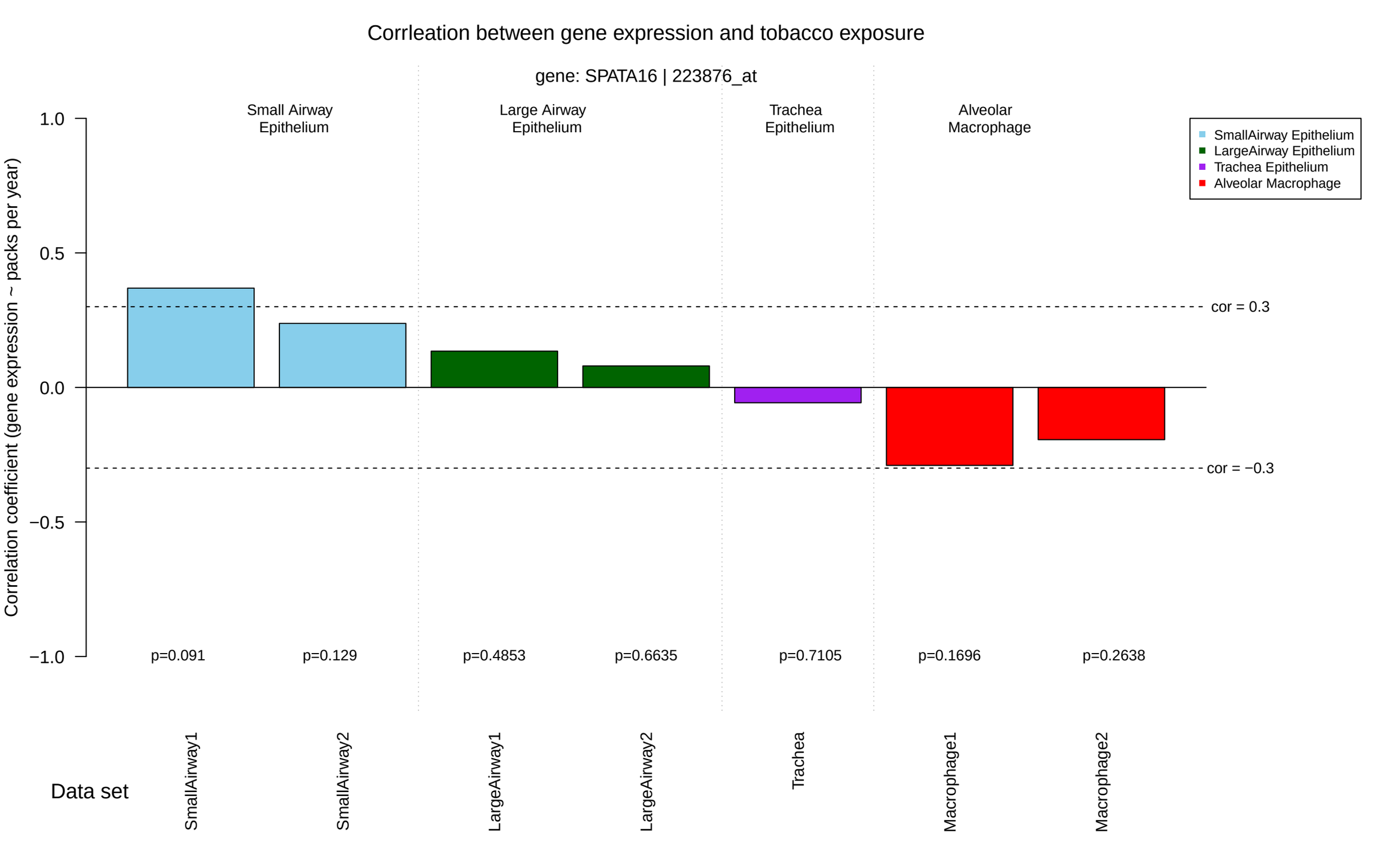

set: LargeAirway1 (p=0.2769)
SPATA16 | 223876_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3189)
SPATA16 | 223876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8409)
SPATA16 | 223876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3385)
SPATA16 | 223876_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0602)
SPATA16 | 223876_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA16 | 223876_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.091
p=0.129
p=0.4853
p=0.6635
p=0.7105
p=0.1696
p=0.2638
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2